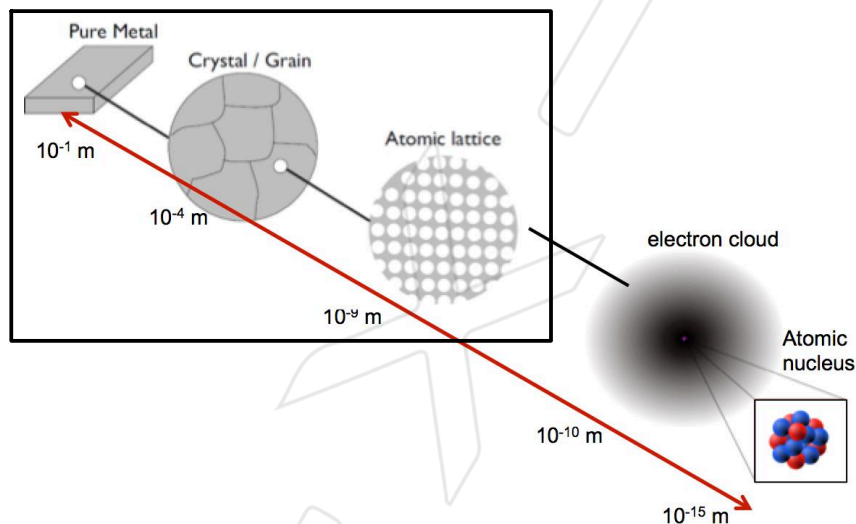


1. Basics: structure of metals

1. Structure sensitive and structure insensitive properties

There are different material properties. Some of the properties are only decided by the atom bonds (e.g. metallic bonds, ionic bonds) and crystal packing (e.g. HCP, BCC and FCC). Those structure insensitive properties are e.g. Young's Modulus, density. Some properties such as yield strength, tensile strength and ductility are also influenced by the material microstructure (e.g. grain size) caused by process parameter etc. so they are structure sensitive properties.

Mark in the figure "Hierarchy of structure" below at which scale level(s), the material properties can be changed by changing the material structures?



2. Some definitions

Polymorphism: pure metal that have different crystal structures at different temperatures

_____, such as Ti and steel.

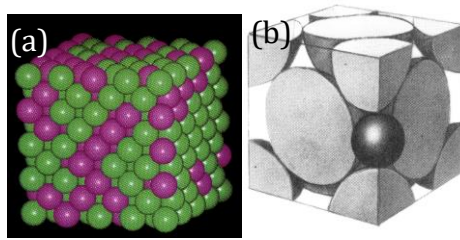
Alloys: mixture of the elements

Phase: a region with some physical and chemical properties

Phase boundary: boundary between two phases

3. Alloy: solid solutions

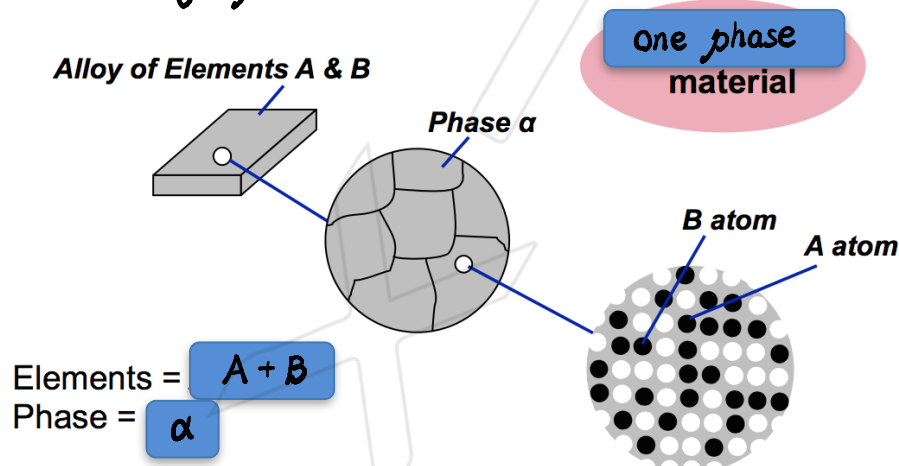
A mixture of elements can form solid solutions which are stronger than pure metals.



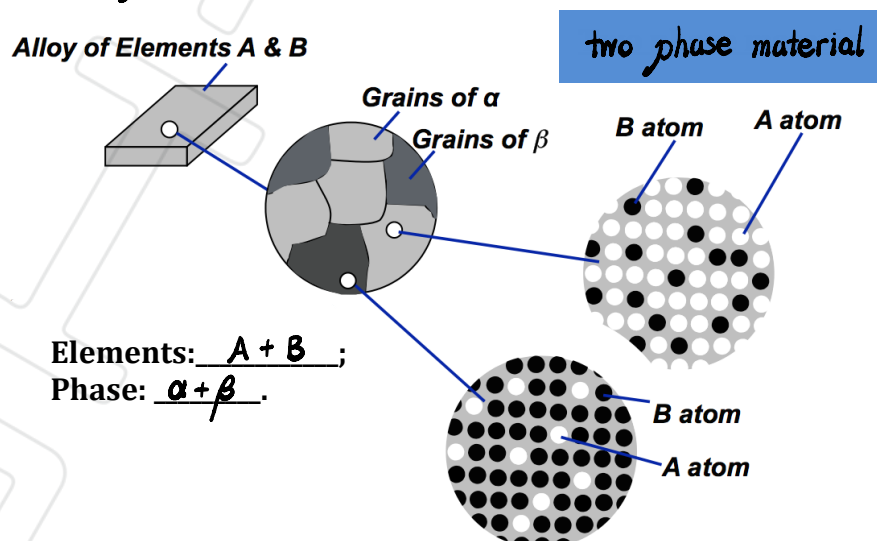
There are different types of solid solutions: substitutional solid solution (Figure a) and interstitial solid solution (Figure b).

4. Solubility limit and phases

In some cases, there is no solubility limit between two elements such as Ni and Cu, so those two elements can be mixed together unlimitly and form a single phase material.



But in many cases, there is a solubility limit and the solute precipitates out as another solution to form a second phase material.



5. Grain boundary and phase boundary

Most single-phase materials are made up of numerous grains. Each grain is a region with the same crystal orientation. Grain Boundaries have an energy associated with them.

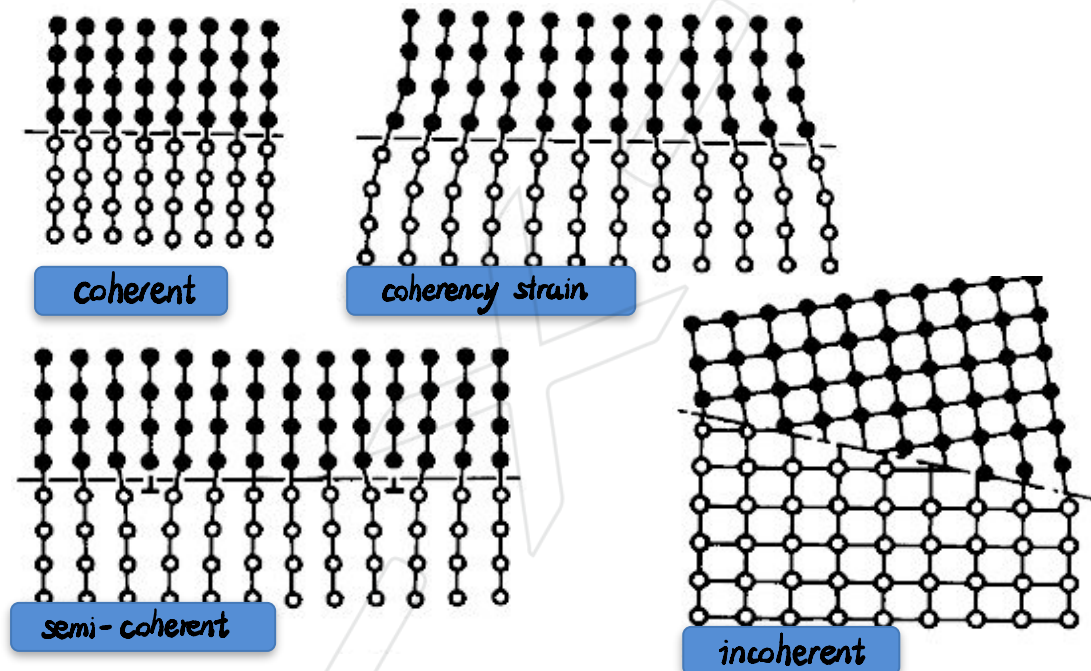
Two-phase materials are also made up of grains: grains of α and grains of β .

Each grain is a region with the same crystal orientation.

Grain boundaries are the boundaries between two different grains and also have an energy associated with them.

Depending on the energy associated with the two phases, there are different boundary type in metals: coherent, coherency strain,

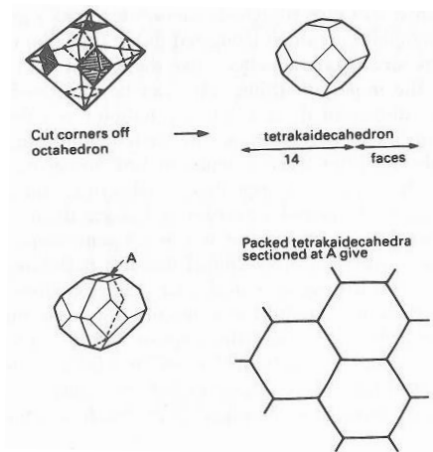
semi-coherent and incoherent; in which coherent has the lowest energy while incoherent has the highest energy.



6. Shape of grains associated with boundary energy.

The shape of grains is usually determined by the shape that minimise the total interfacial energy. $E = \gamma_{gb} \cdot A$ (γ_{gb} is grain boundary energy, A is interfacial area). When γ_{gb} is isotropic (does not depend on grain boundary orientation angle), then the lowest interfacial energy is that with the lowest interfacial area.

In a single phase material, grain tends to have a tetrakaidecahedron shape to minimise the interfacial area.

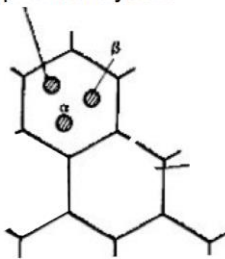


For two-phase material

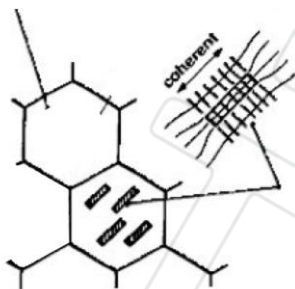
- Consider phase β forming within a grain of phase α :

--- If $\gamma_{\alpha\beta}$ is isotropic, $\gamma_{\alpha\beta}$ does not depend on α or β orientation. The lowest energy interface occurs for the shape with the lowest area, leading to a sphere of phase β in phase α .

Isotropic $\gamma_{\alpha\beta}$ gives spherical crystals



--- If $\gamma_{\alpha\beta}$ is non-isotropic, $\gamma_{\alpha\beta}$ depends on orientation between α and β . The lowest energy interface occurs for the shape with the lowest total $E = \gamma_{\alpha\beta} A$. If one interface has a lower energy, phase β will form a plate in phase α .



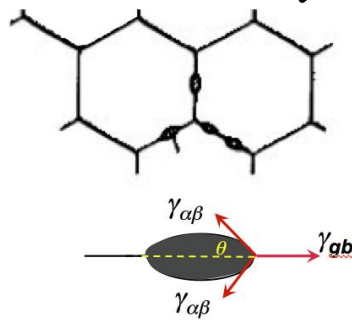
- Considering the case when interphase boundaries meet:

--- In the case that β forms at α grain boundaries:

The β particle shape depends on the interfacial energies. β meets the α grain boundary at an angle of θ , where

$$2\gamma_{\alpha\beta} \cos \theta = \gamma_{gb}$$

This is a balance of surface tensions.



--- Special case:

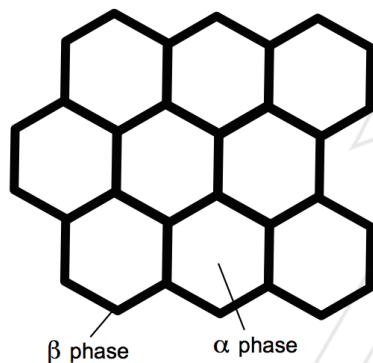
If $\gamma_{\alpha\beta} \leq \frac{1}{2} \gamma_{gb}$

Then $\cos \theta = 1$

So $\theta = 0$

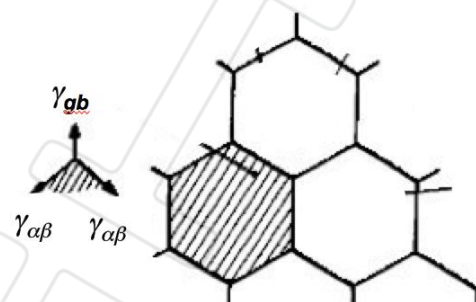
$$2\gamma_{\alpha\beta} \cos \theta = \gamma_{gb}$$

β phase will spread along all α grain boundaries causing interesting microstructure and material properties (such as Gallium Induced Structural Failure of an Aluminum Sheet).

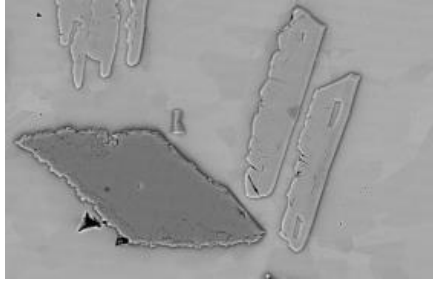


--- In some cases β can form their own grains adjacent to the α grains.

- If $\gamma_{\alpha\beta} = \gamma_{gb}$, the β grain shape will be also tetrakaidecahedral.



- Usually $\gamma_{\alpha\beta} \neq \gamma_{gb}$ and then complex β grain shapes occur. The β grain takes a shape that minimise the interfacial energy.



7. Importance of processing route

Metallic microstructures are influenced by alloy constitution and geometry of the microstructure. Alloy constitution includes factors such as the number of phase present, the volume fraction of each phase and the composition of each phase and those factors are depending on their overall composition and temperature. The geometry of the microstructure are influenced by the size and spacing of each phase and the shape of each phase, which are also effected by processing route. Processing route plays an important role to the material microstructure and therefore their properties.